

Group By Clause in SQL Server with Examples

In this article, I am going to discuss the **Group by Clause in SQL Server** with examples. Please read our previous article where we discussed the [Top n Clause in SQL Server](#) in detail.

Group By Clause in SQL Server

The Group by Clause in SQL Server is used to divide similar types of records or data as a group and then return. If we use group by clause in the query then we should use grouping/aggregate function such as count(), sum(), max(), min(), and avg() functions.

When we implement group by clause first the data of the table will be divided into the separate group as per the column and later aggregate function will execute on each group data to get the result. That means first Group By clause is used to divide similar types of data as a group and then an aggregate function is applied to each group to get the required results.

Syntax of Group By Clause in SQL Server

```
SELECT expression1, expression2, expression_n,  
       aggregate_function (expression)  
FROM tables  
[WHERE conditions]  
GROUP BY expression1, expression2, expression_n;
```

Parameters or Arguments used in Group By Clause in SQL Server:

1. **expression1, expression2, expression_n:** The expressions that are not encapsulated within an aggregate function must be included in the GROUP BY clause.
2. **aggregate_function:** The aggregate function is nothing but such as SUM, COUNT, MIN, MAX, or AVG functions that we should use while we are using the Group by Clause in SQL Server.
3. **Tables:** Tables are nothing but the name of the table or tables from which we want to retrieve the data.
4. **WHERE conditions:** It is optional. If you want to retrieve the data based on some conditions then you need to specify such conditions using the Where Clause in SQL Server.

Example to Understand Group By Clause in SQL Server

Let us understand SQL Server Group By Clause with Examples. We are going to use the following Employee table to understand the Group By Clause in SQL Server with Examples.

ID	Name	EmailID	Gender	Department	Salary	Age	CITY
1	PRANAYA	PRANAYA@G.COM	Male	IT	25000	30	MUMBAI
2	TARUN	TARUN@G.COM	Male	Payroll	30000	27	ODISHA
3	PRIYANKA	PRIYANKA@G.COM	Female	IT	27000	25	BANGALORE
4	PREETY	PREETY@G.COM	Female	HR	35000	26	BANGALORE
5	RAMESH	RAMESH@G.COM	Male	IT	26000	27	MUMBAI
6	PRAMOD	PRAMOD@G.COM	Male	HR	29000	28	ODISHA
7	ANURAG	ANURAG@G.COM	Male	Payroll	27000	26	ODISHA
8	HINA	HINA@G.COM	Female	HR	26000	30	MUMBAI
9	SAMBIT	HINA@G.COM	Male	Payroll	30000	25	ODISHA
10	MANOJ	MANOJ@G.COM	Male	HR	30000	28	ODISHA
11	SWAPNA	SWAPNA@G.COM	Female	Payroll	28000	27	MUMBAI
12	LIMA	LIMA@G.COM	Female	HR	30000	30	BANGALORE
13	DIPAK	DIPAK@G.COM	Male	Payroll	32000	25	BANGALORE

Please use the below SQL Script to create and populate the Employee table with sample data that we are going to use in this article.

```
-- Create Employee table
CREATE TABLE Employee
(
    ID INT PRIMARY KEY IDENTITY(1,1),
    Name VARCHAR(100),
    EmailID VARCHAR(100),
    Gender VARCHAR(100),
    Department VARCHAR(100),
    Salary INT,
    Age INT,
    CITY VARCHAR(100)
)
GO
--Insert data into Employee table
INSERT INTO Employee VALUES('PRANAYA','PRANAYA@G.COM','Male','IT', 25000, 30, 'MUMBAI')
INSERT INTO Employee VALUES('TARUN','TARUN@G.COM','Male','Payroll', 30000, 27, 'ODISHA')
INSERT INTO Employee VALUES('PRIYANKA','PRIYANKA@G.COM','Female','IT', 27000, 25, 'BANGALORE')
INSERT INTO Employee VALUES('PREETY','PREETY@G.COM','Female','HR', 35000, 26, 'BANGALORE')
INSERT INTO Employee VALUES('RAMESH','RAMESH@G.COM','Male','IT', 26000, 27, 'MUMBAI')
INSERT INTO Employee VALUES('PRAMOD','PRAMOD@G.COM','Male','HR', 29000, 28, 'ODISHA')
INSERT INTO Employee VALUES('ANURAG','ANURAG@G.COM','Male','Payroll', 27000, 26, 'ODISHA')
INSERT INTO Employee VALUES('HINA','HINA@G.COM','Female','HR', 26000, 30, 'MUMBAI')
INSERT INTO Employee VALUES('SAMBIT','HINA@G.COM','Male','Payroll', 30000, 25, 'ODISHA')
INSERT INTO Employee VALUES('MANOJ','MANOJ@G.COM','Male','HR', 30000, 28, 'ODISHA')
INSERT INTO Employee VALUES('SWAPNA','SWAPNA@G.COM','Female','Payroll', 28000, 27, 'MUMBAI')
INSERT INTO Employee VALUES('LIMA','LIMA@G.COM','Female','HR', 30000, 30, 'BANGALORE')
```

```
INSERT INTO Employee VALUES('DIPAK','DIPAK@G.COM','Male','Payroll', 32000,
25,'BANGALORE')
GO
```

WAQ to find total employees in the organization.

SELECT COUNT(*) AS TotalEmployee FROM Employee

TotalEmployee
13

WAQ to find the number of employees working in each department in the company.

Here, we need to group the employees by department, and then we need to apply the count function to each group. The following SQL query exactly does the same.

```
SELECT Department, COUNT(*) AS TotalEmployee
FROM Employee
GROUP BY Department
```

When we execute the above query, it gives us the below output.

Department	TotalEmployee
HR	5
IT	3
Payroll	5

Note: If we use Group By Clause in a query, first the data in the table will be divided into different groups based on the column specified in the group by clause and then execute the aggregate function on each group to get the results.

WAQ to find the total salary in each department of the organization.

```
SELECT Department, TotalSalary = SUM(Salary)
FROM Employee
GROUP BY Department
```

When we execute the above query, it gives us the below output.

Department	TotalSalary
HR	150000
IT	78000
Payroll	147000

WAQ to find the highest salary in each department in the organization.

```
SELECT Department, MaxSalary = MAX(SALARY)
FROM Employee
GROUP BY Department
```

When we execute the above query, it gives us the below output.

Department	MaxSalary
HR	35000
IT	27000
Payroll	32000

While working with the group by clause in a query we need to follow or consider the following

1. When the aggregate function is applied to a group it returns only a single value but each group can return a value.
2. Use Group By clause only on a column that contains duplicate values, never applies it on unique columns.

Write a query to get the number of employees working in each Gender per department.

```
SELECT Department, Gender, EmployeeCount = COUNT(*)
FROM Employee
GROUP BY Department, Gender
ORDER BY Department
```

When we execute the above query, it gives us the below output.

Department	Gender	EmployeeCount
HR	Female	3
HR	Male	2
IT	Female	1
IT	Male	2
Payroll	Female	1
Payroll	Male	4

When we use multiple columns in a group by clause first data in the table is divided based on the first column of the group by clause and then each group is subdivided based on the second column of the group by clause and then the group function is applied on each inner group to get the result.

Write a query to get the highest salary for the organization.

SELECT MAX(Salary) as MaxSalary FROM Employee

MaxSalary
35000

Write a Query for retrieving total salaries by the city.

We are applying the SUM() aggregate function on the Salary column, and grouping by city column. This effectively adds all the salaries of employees within the same city.

```
SELECT CITY, SUM(Salary) as TotalSalary
FROM Employee
GROUP BY CITY
```

When we execute the above query, it gives us the following output.

CITY	TotalSalary
BANGALORE	124000
MUMBAI	105000
ODISHA	146000

Note: If you omit, the group by clause and try to execute the query, you get an error –

Column 'Employee.CITY' is invalid in the select list because it is not contained in either an aggregate function or the GROUP BY clause.

Write a Query for retrieving total salaries by city and by gender.

It's possible to group by multiple columns. In this query, we are grouping first by city and then by gender.

```
SELECT CITY, Gender, SUM(Salary) as TotalSalary
FROM Employee
GROUP BY CITY, Gender
```

When we execute the above query, it gives us the below output.

CITY	Gender	TotalSalary
BANGALORE	Female	92000
MUMBAI	Female	54000
BANGALORE	Male	32000
MUMBAI	Male	51000
ODISHA	Male	146000

Write a Query for retrieving total salaries and the total number of employees by City, and by gender.

The only difference here is that, we are using Count() aggregate function.

```
SELECT City, Gender, SUM(Salary) as TotalSalary,
COUNT(ID) as TotalEmployees
FROM Employee
```

GROUP BY CITY, Gender

City	Gender	TotalSalary	TotalEmployees
BANGALORE	Female	92000	3
MUMBAI	Female	54000	2
BANGALORE	Male	32000	1
MUMBAI	Male	51000	2
ODISHA	Male	146000	5

Write a query to find out the highest salary of each department along with the name of the employee.

```
SELECT Department, MAX(Salary) as Salary, Name
FROM Employee
GROUP BY Department - Invalid
```

It will give the error as **Column 'Employee.Name' is invalid in the select list because it is not contained in either an aggregate function or the GROUP BY clause.**

Note: The above query will not be executed because using group by clause in a statement the select list can contain only three things

1. Columns that are associated with the GROUP BY clause.
2. Aggregate or GROUP functions
3. Constants

In our above query Name column does not fall under any of the three so it cannot be used in the select list.

```
SELECT Department, MAX(Salary) as Salary, Name
FROM Employee
GROUP BY Department -- Invalid
SELECT Department, MAX(SALARY), GETDATE()
FROM Employee
GROUP BY Department --Valid
SELECT Department, Gender, MAX(Salary), 'Hello' as FixedValue
FROM Employee
GROUP BY Department, Gender --Valid
```

In the next article, I am going to discuss [Having Clause in SQL Server](#) with examples. Here, in this article, I try to explain **Group By Clause in SQL Server** with some examples. I hope this article will help you with your needs. I would like to have your feedback. Please post your feedback, question, or comments about this article.